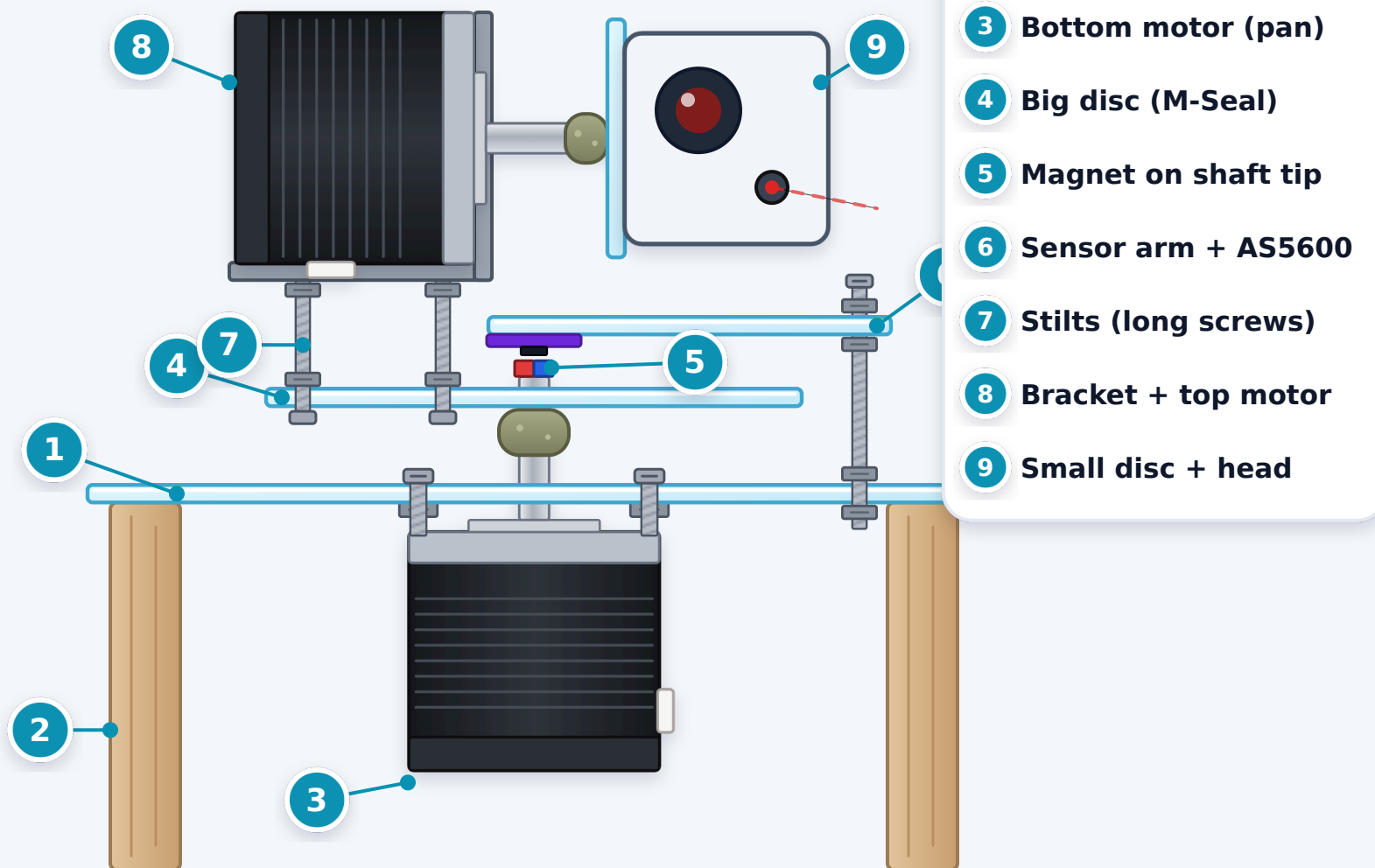




This is what you are building

Side view of the finished MK2

bottom motor turns left-right · top motor nods up-down



- 1 Two motors. The **bottom** one turns left-right. The **top** one nods up-down.
- 2 The camera and the laser sit together in the **head**, so the camera sees its own laser dot.
- 3 Flat acrylic, screws and M-Seal only. **No 3D printing, no coupler.**



Tonight's goal: build this, wire it, and watch it lock onto the blinking beacon on the website.

0

PART 1 · THE SPINNING BASE

What you have, and the only 3 things to get



2 motors
(NEMA17)



2 motor
drivers



2 Arduino
Nano



2 sensors
+ magnets



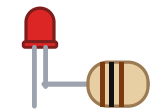
Multiplexer
board



Laser
(KY-008)



USB webcam
(arriving)



LEDs
+ resistors



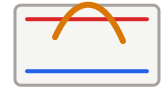
Acrylic
sheets



Screws
+ nuts



12 V adapter
+ DC socket



Breadboard
+ wires



Ping-pong ball
+ AA holder



Glue gun
+ solder



Multimeter



Cable ties



GET FROM ANY HARDWARE / GENERAL SHOP (if not at home)



M-Seal or Araldite

₹50-100 · glues the discs



3 AA cells

₹30 · the decoy



Fevikwik

₹10-20 · the magnet

- 1 Everything with a green tick you **already have**. The webcam can come later — use the laptop camera till then.
- 2 The orange box: any hardware, paint or general store has these. **M-Seal is the one thing you can't skip.**
- 3 No coupler, no red acrylic, no belt. Keep the yellow part as a spare; we don't need it.



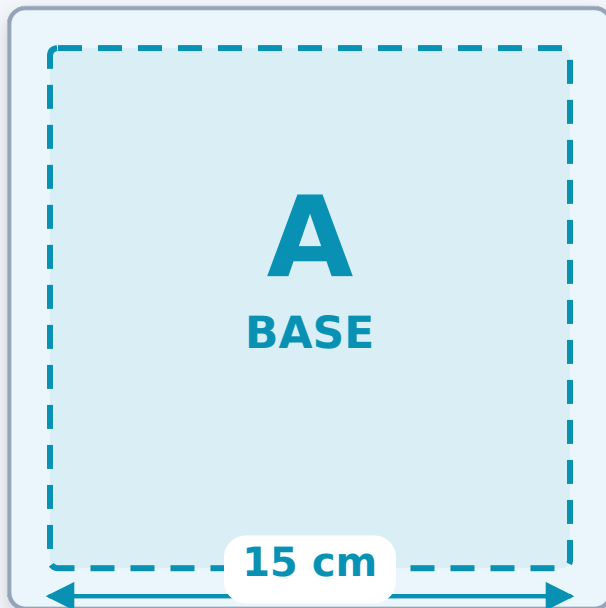
Nothing else to buy. Everything in this guide uses only these parts.

1

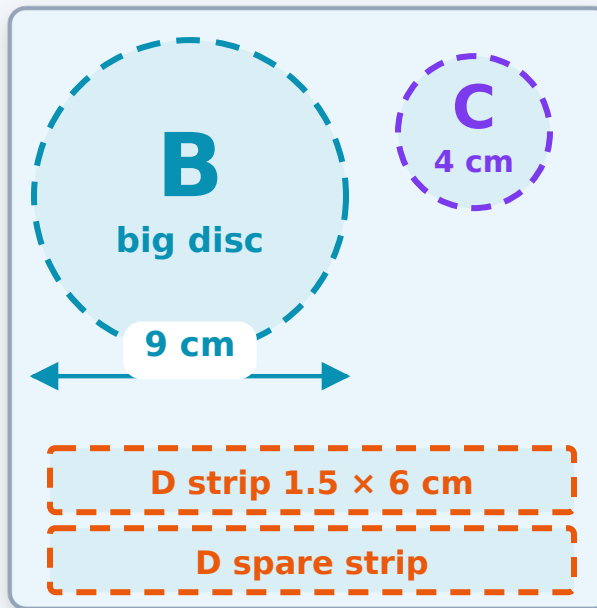
PART 1 · THE SPINNING BASE

Cut 4 pieces from the acrylic

SHEET 1



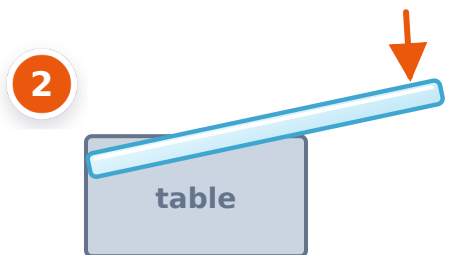
SHEET 2



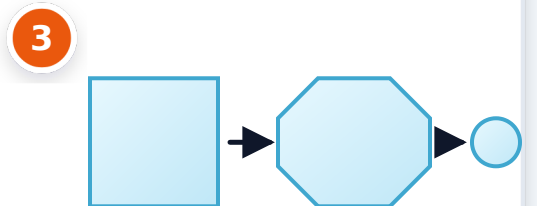
HOW TO CUT



Scratch the line
8-10 times with a
cutter, along a ruler



Line on the table
edge. Press down
hard: it snaps clean.



Square → cut corners
→ sand it round.

YOU NEED

- | | |
|-----------------------|-------------------|
| A Base | 15 × 15 cm square |
| B Big disc | 9 cm circle |
| C Small disc | 4 cm circle |
| D Sensor strip | 1.5 × 6 cm |

Keep the paper film on while cutting — no scratches.

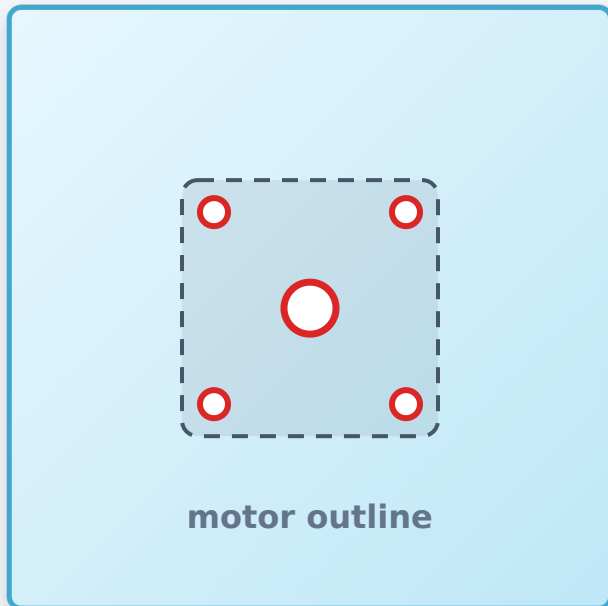
- 1 Straight cuts:** scratch the line 8-10 times with a cutter along a steel ruler, then snap it on the table edge.
- 2 Circles:** draw round a cup or lid. Cut a square, cut off the corners, then sand it round. It does not need to be perfect.
- 3** Peel the paper film off only at the very end.



4 pieces ready: base, big disc, small disc, one sensor strip (plus a spare).

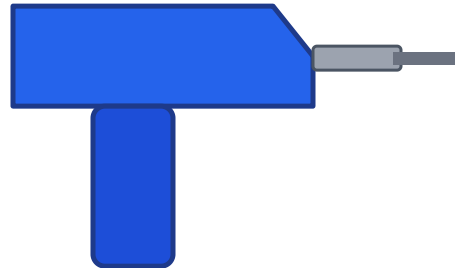
Make the holes

MARK THEM



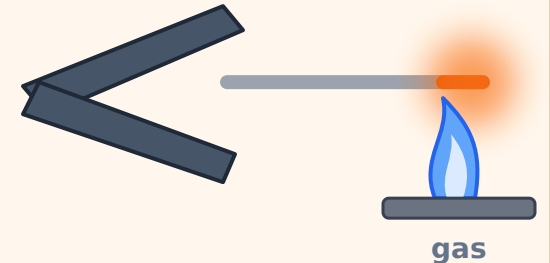
Motor face-down in the middle. Dot through its 4 holes, and the middle.

WITH A DRILL



Small tape on the spot so the bit does not slide. Go SLOW, no pressure.

NO DRILL? HOT NAIL



Hold a nail with pliers. Heat the tip on the gas 20 seconds. Push it through the dot.

Never touch the tip!

WHICH HOLES

- A Base** 4 small (screws) + 1 big middle hole (8 mm, the shaft must NOT touch)
- B Big disc** 1 middle hole = exactly the shaft size (tight) + 3 small holes on ONE side
- C Small disc** 1 middle hole = exactly the shaft size (tight) + 2 small holes
- D Strip** 1 small hole at one end

- 1 Use the motor itself as the stencil — then the holes always match.
- 2 Middle holes of the **discs** must be **tight** on the motor shaft. Start small, widen slowly by twisting a screwdriver in it.
- 3 The **base** middle hole is big on purpose: the shaft spins inside it without touching.



Push a disc on the motor shaft: it goes on snug, not loose and not forced.

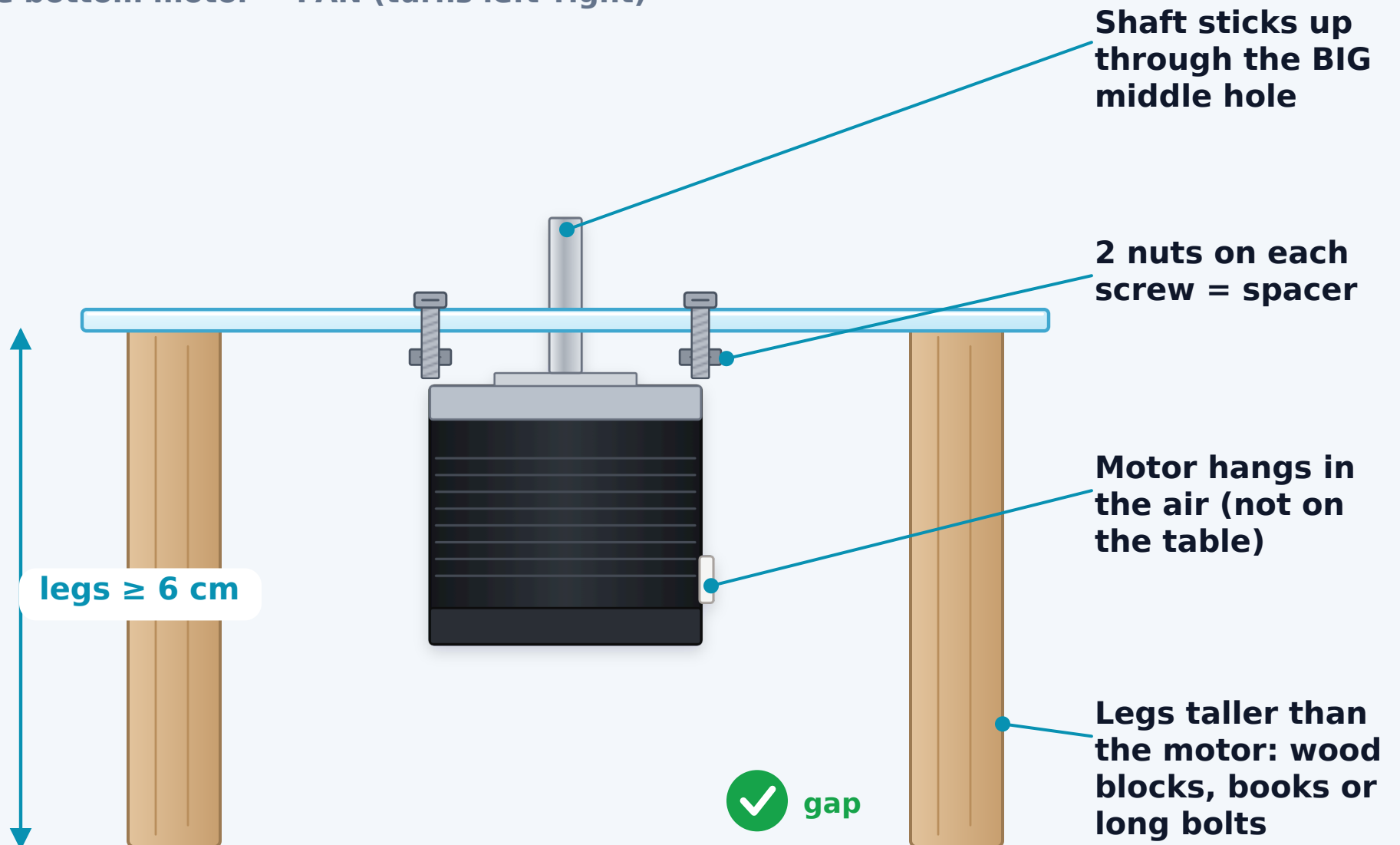
3

PART 1 · THE SPINNING BASE

Base, legs and the bottom motor

Side view

the bottom motor = PAN (turns left-right)



- 1 Motor goes **under** the base, shaft pointing **up** through the big middle hole.
- 2 4 screws from the top. Put **2 nuts on each screw** between the base and the motor, so the round bump on the motor does not press on the acrylic.
- 3 Stand the base on 4 legs **taller than the motor**, so the motor hangs free.



The base sits flat and does not rock. The motor does not touch the table.

Glue the big disc onto the shaft

Close-up, side view

the disc is glued straight onto the metal shaft

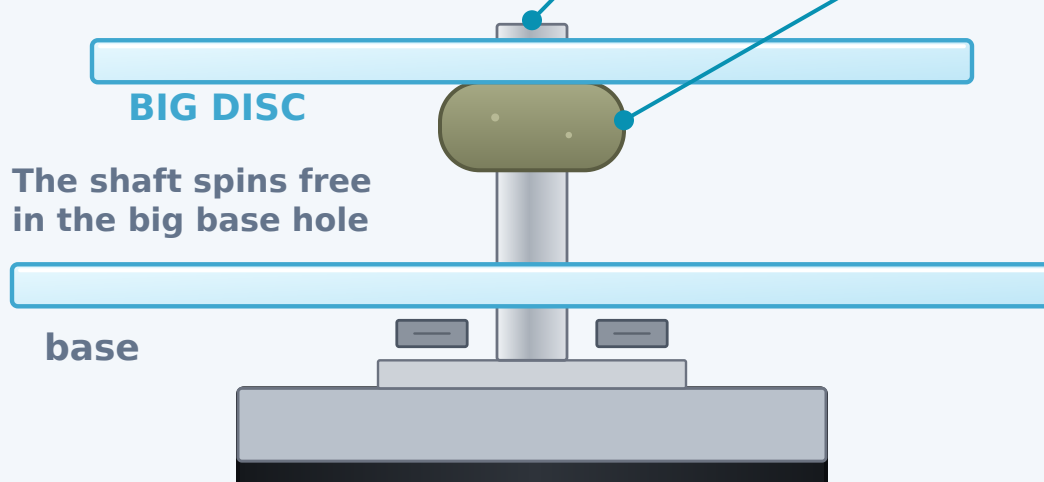
KEEP IT FLAT



Phone level app on the disc: bubble in the middle = flat.

Shaft tip sticks out about 2 mm — keep it **CLEAN** (magnet goes here next)

Fat M-Seal ring **UNDER** the disc, all round the shaft



MIX M-SEAL



Equal bits of both colours. Roll till it is **ONE** colour.

Hard in about 1 hour.

- 1 Push the big disc onto the shaft until only about **2 mm** of shaft sticks out on top.
- 2 Press a fat ring of mixed M-Seal **under** the disc, all round the shaft. Keep the **top of the shaft clean**: the magnet goes there.
- 3 Keep the disc flat (phone level app). **Don't turn it for 1 hour** while it hardens.



After 1 hour: spin it by hand. It turns flat, no wobble, and the disc never slips on the shaft.

Magnet and the sensor arm

Close-up, side view



Magnet test: N and S
on the SIDES (half red,
half blue) = correct

Magnet glued on
the TIP of the shaft
(Fevikwik)

Sensor chip faces
DOWN, exactly over
the magnet

Long screw = post.
It stands on the base
and **NEVER** turns.

Nuts above and
below the strip set
the height

2 credit cards fill the gap —
slide them out after

base

- 1 Put one drop of Fevickwik on the **very tip of the shaft** and stick the small round magnet there.
- 2 Stand a long screw on the base, 4–5 cm from the middle, locked with nuts. The acrylic strip sits on it and reaches over the magnet.
- 3 Stick the sensor (AS5600) under the strip's end, **chip facing down** onto the magnet. Gap = 2 credit cards.



Spin the disc: the strip and the sensor never move and never touch anything.

Stilts, bracket and the top motor

Side view of the top half

Top motor = TILT
(nods up-down)

L-bracket

3 long screws =
stilts, on the side
AWAY from the
sensor arm

shaft points
SIDWAYS

gap

The bracket must pass
OVER the sensor arm with
a gap. Turn the nuts on
the stilts to raise it.

- 1 Push 3 long screws up through the big disc's small holes — the side **away from** the sensor arm. Lock each with a nut.
- 2 Screw the L-bracket's flat part onto the stilts, with nuts above and below it.
- 3 Bolt the second motor to the bracket's standing part with 4 screws, **shaft pointing sideways**.



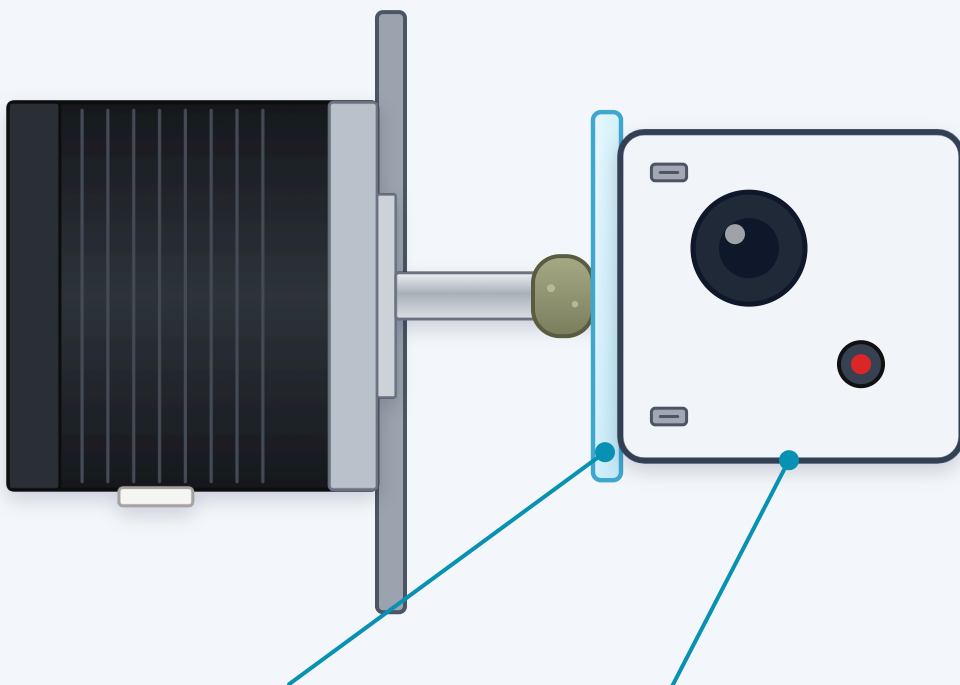
Turn the big disc a quarter turn left and right by hand: the bracket never hits the sensor arm or its screw.

7

PART 2 · THE TOP AND THE HEAD

Small disc and the head

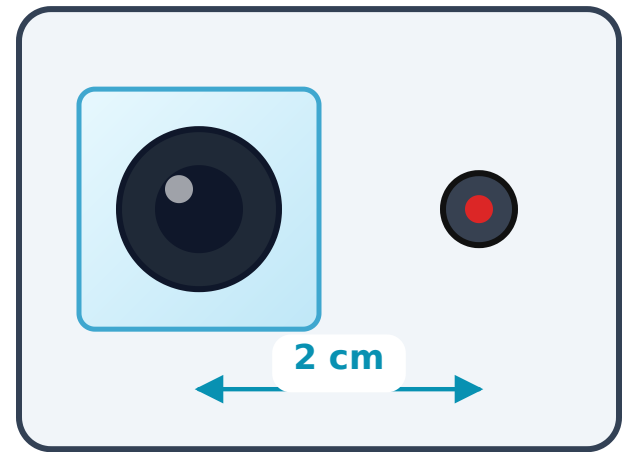
Side view of the head



Small disc glued on the top shaft (same M-Seal trick)

Head box screwed **FLAT** on the small disc, centred → it balances

FRONT OF THE HEAD



camera behind clear window

laser

Both look the **SAME** way. Hot-glue them.

The 2nd sensor is **OPTIONAL**

The top motor works without it. Skip it tonight. The code now works with just the bottom sensor.

Leave the cables loose, tie them with cable ties: slack at every joint.

- 1 Glue the small disc on the top motor's shaft the same way: M-Seal, 2 mm sticking out, wait 1 hour.
- 2 Screw the head box **flat onto the small disc**, centred. Centred = balanced, so the motor does not strain.
- 3 Inside: webcam behind the clear window, laser beside it (2 cm), both looking the same way. Hot-glue both.



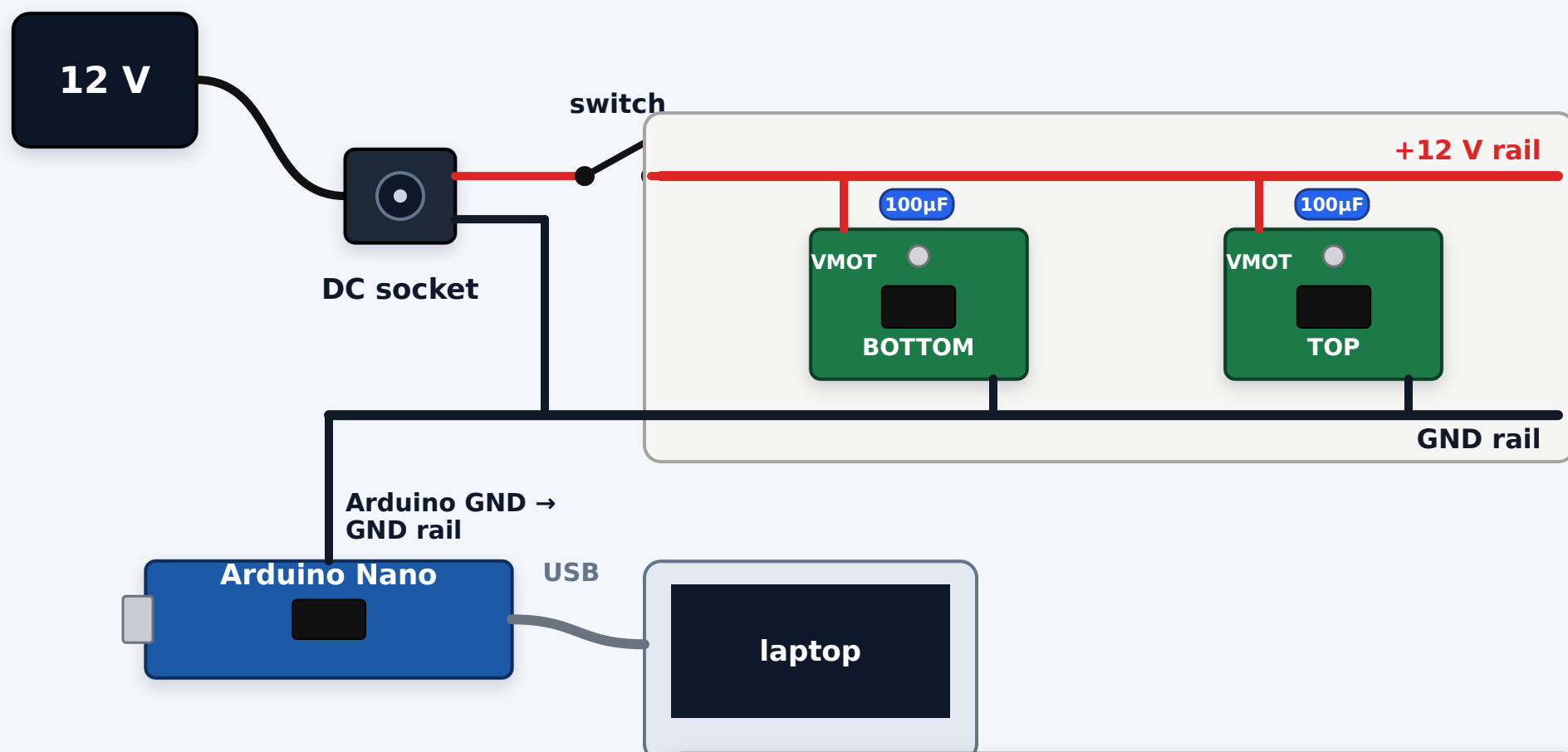
On a wall 2 m away the laser dot lands near the middle of the camera picture.

8

PART 3 · WIRES AND CODE

Power: 12 V for the motors only

Two powers, ONE ground



Find + on the socket:
adapter in, multimeter
on DC V. The 2 pins
that read 12 V: red
probe = +. Mark it.

- 12 V goes **ONLY** to the motor drivers (VMOT)
- Never pull a motor wire while the switch is ON
- Capacitor: long leg to +12 V, stripe to GND
- Arduino GND to the GND rail — one shared ground

- 1 12 V adapter** → DC socket → switch → the **red** rail. Socket — → the **blue** (GND) rail.
- Each driver: VMOT to red, GND to blue, and a 100 µF capacitor across them (stripe to GND).
- The **laptop USB** powers the Arduino. One wire joins the Arduino GND to the GND rail.



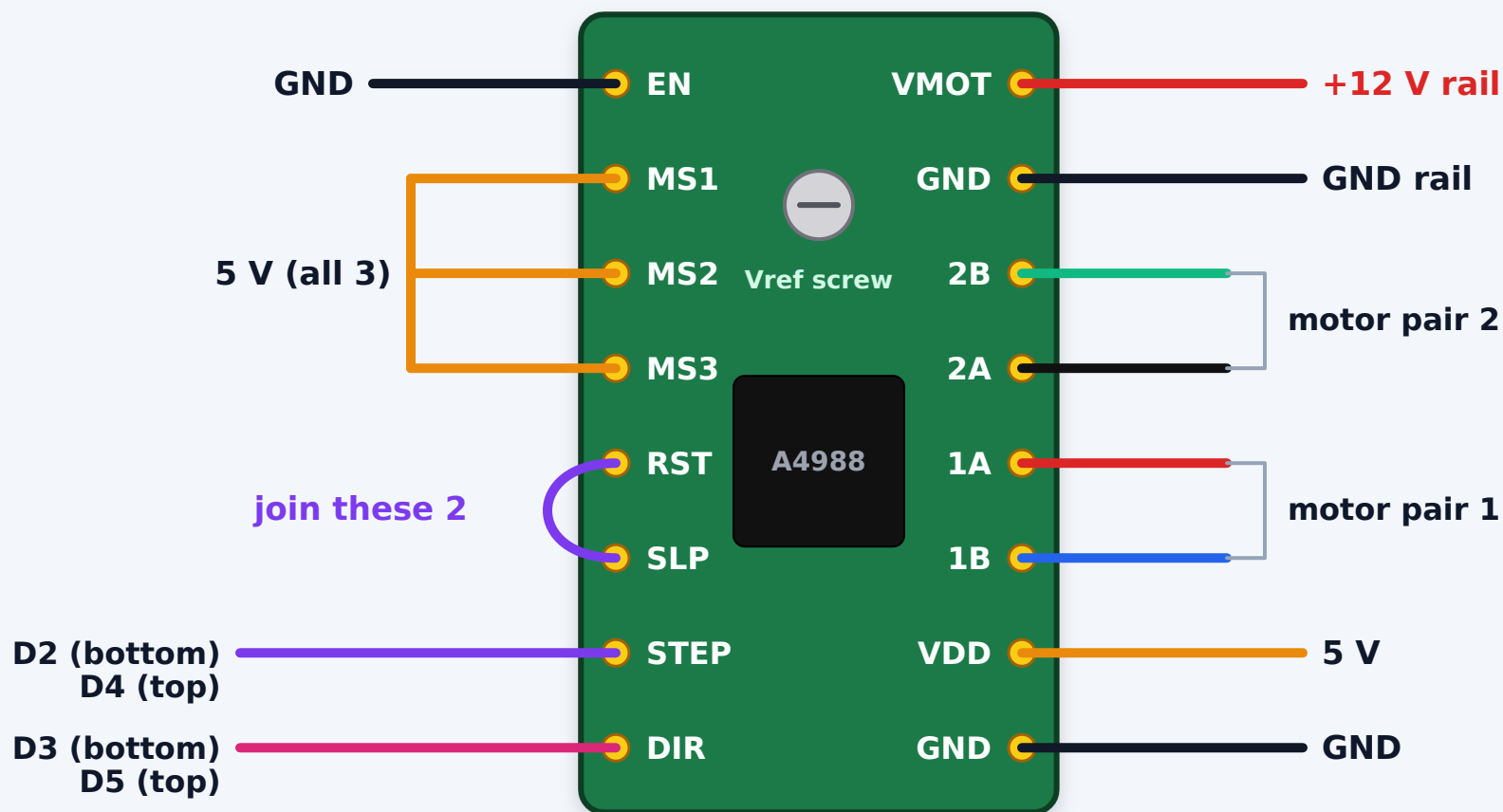
Switch ON (nothing else plugged in yet): the red rail reads about 12 V on the multimeter.

9

PART 3 · WIRES AND CODE

Wire a motor driver (do it twice)

One driver · wire it the same way twice



Find the 2 pairs

Multimeter on Ω : the 2 wires that read 1-4 Ω = one pair. Usually black+green and red+blue. Never mix the pairs.

Vref (USB only, 12 V OFF)

Black probe GND, red probe on the little screw. Turn gently to 0.55 V. Do both drivers.

- 1 Small jumpers: **RST↔SLP** together, **MS1 MS2 MS3** to 5 V, **EN** to GND.
- 2 **STEP** / **DIR** to the Arduino: bottom motor **D2** / **D3**, top motor **D4** / **D5**.
- 3 Motor: one pair on **1A-1B**, the other pair on **2A-2B**. Set Vref to 0.55 V.

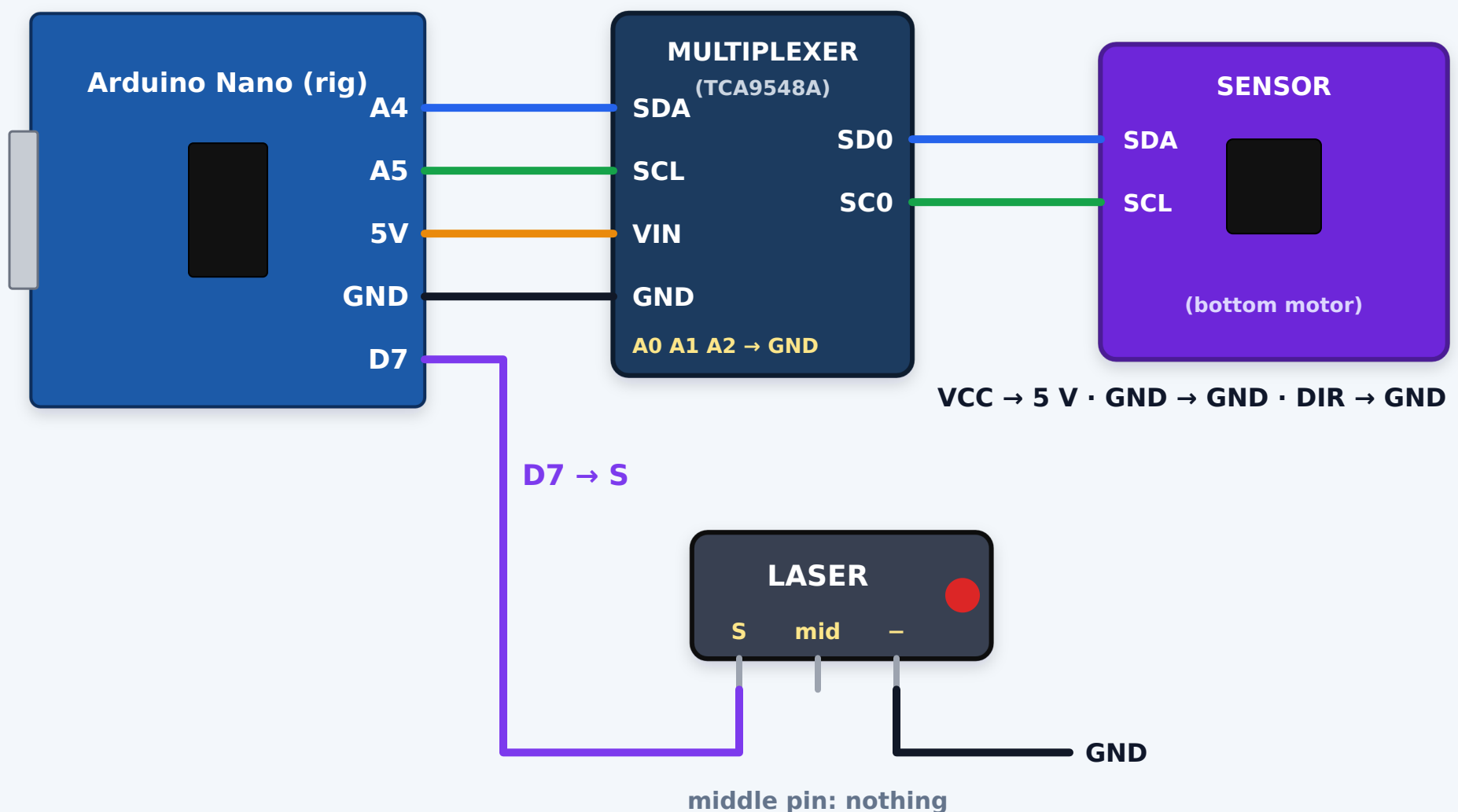


Switch ON: both motor shafts go stiff (hard to turn by hand). The drivers get only warm.

Sensor and laser wires

Sensor through the multiplexer, laser straight in

blue = SDA · green = SCL · orange = 5 V · black = GND



- 1 Arduino **A4→SDA**, **A5→SCL**, 5 V→VIN, GND→GND on the multiplexer. Its A0 A1 A2 → GND.
- 2 Sensor on channel **0** (SD0, SC0). Sensor VCC→5 V, GND→GND, DIR→GND.
- 3 Laser: **S→D7**, **-→GND**. Leave the middle pin empty.



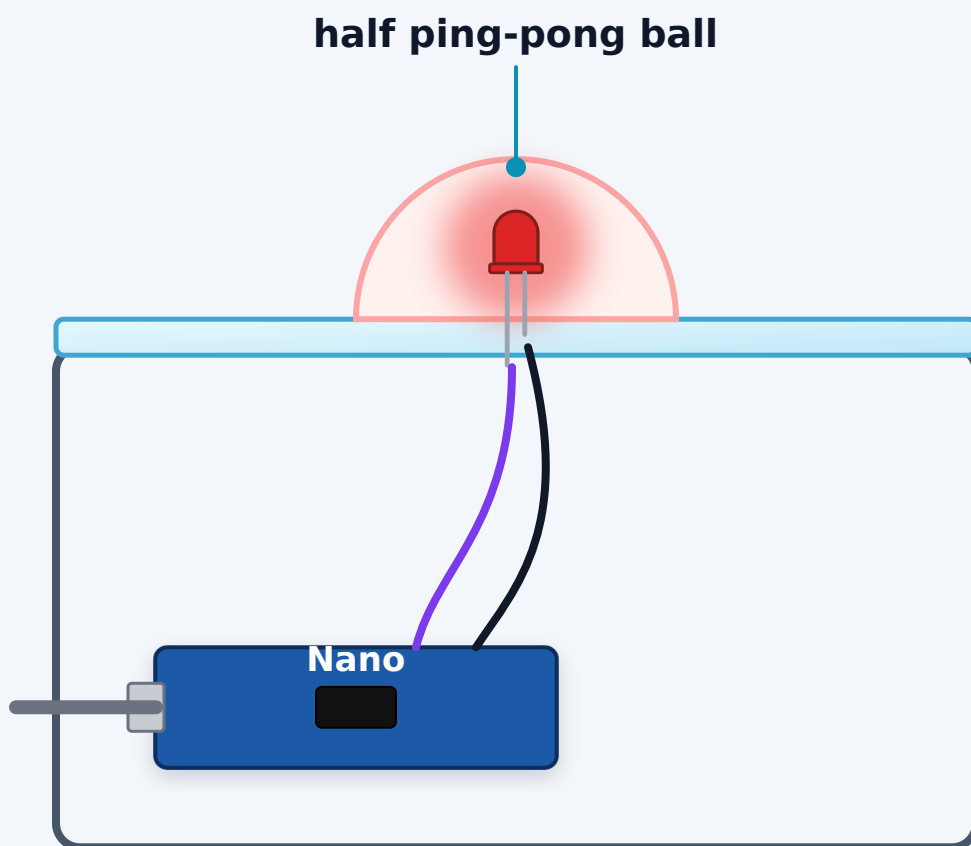
After uploading the code (card 13) the Arduino says encoders=pan. That means the sensor works.

11

PART 4 · TARGETS AND GO LIVE

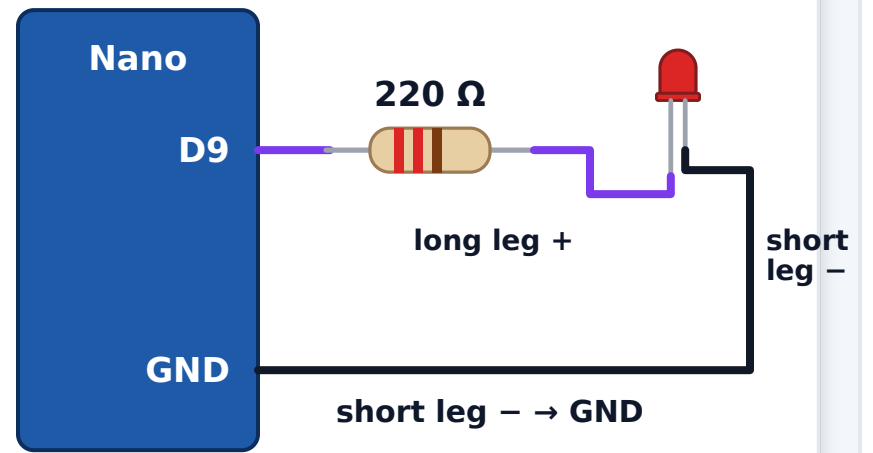
The beacon (the blinking target)

Cut through the middle



USB cable out of a side hole →
phone charger or power bank

THE CIRCUIT



red-red-brown = 220 Ω

BLINKS 4 TIMES A SECOND



Upload **tools/rig/beacon_firmware.ino**
to the **SECOND** Nano.

Slow-motion video: count 4 flashes / s.

- 1 LED **long leg** → 220 Ω → Nano **D9**. LED **short leg** → Nano **GND**.
- 2 Upload **beacon_firmware.ino** to the second Nano. The LED blinks 4 times a second.
- 3 LED through a hole in the lid, half a ping-pong ball glued over it. Power from a phone charger or power bank.

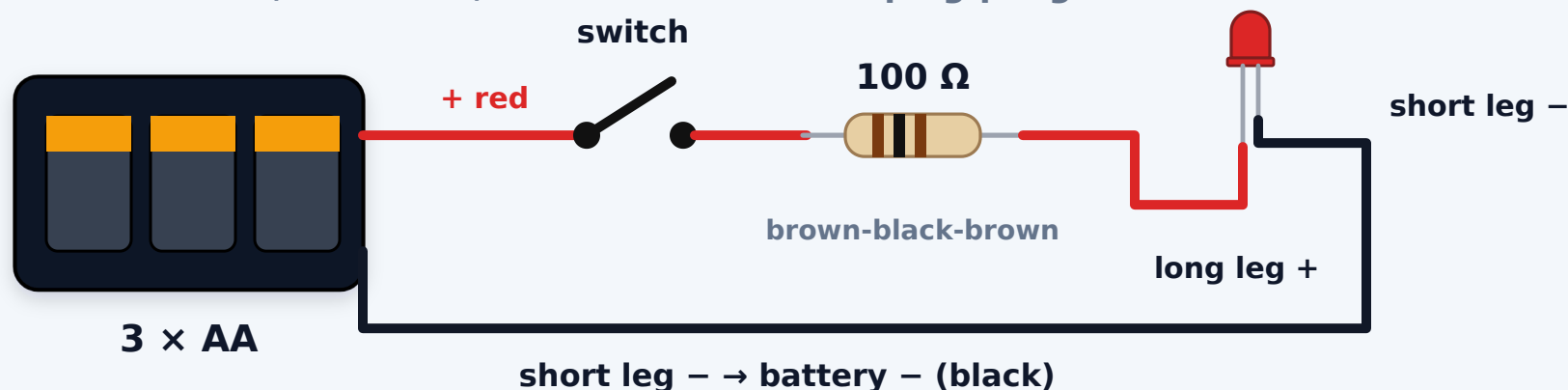


LED blinks steadily. No light? The LED is the wrong way round — swap its two wires.

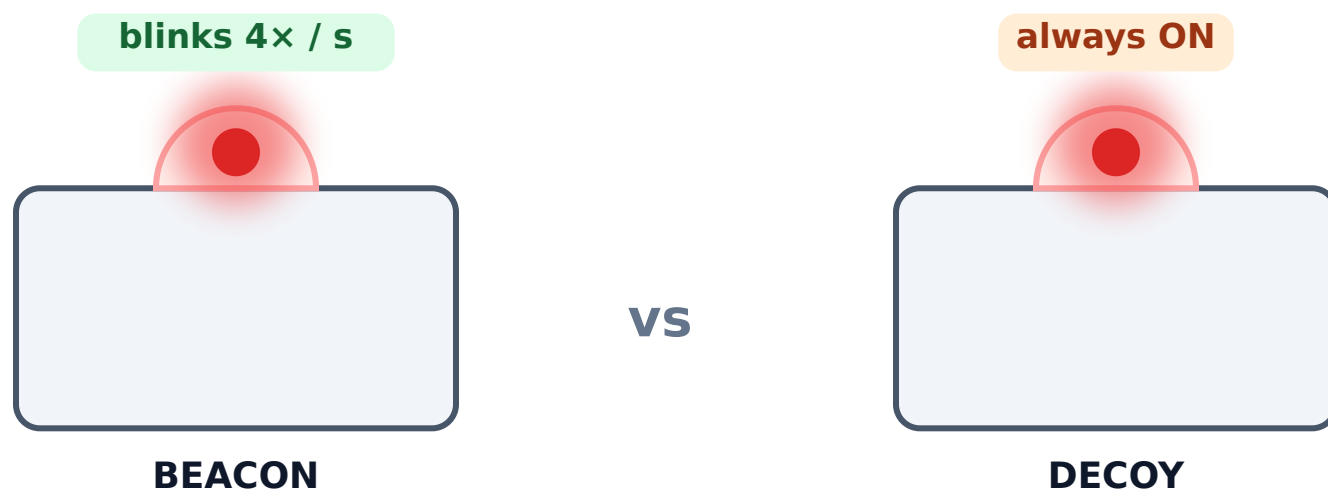
The decoy (the trick light)

The decoy: no Arduino, no code

Same red LED, same box, the other half of the ping-pong ball



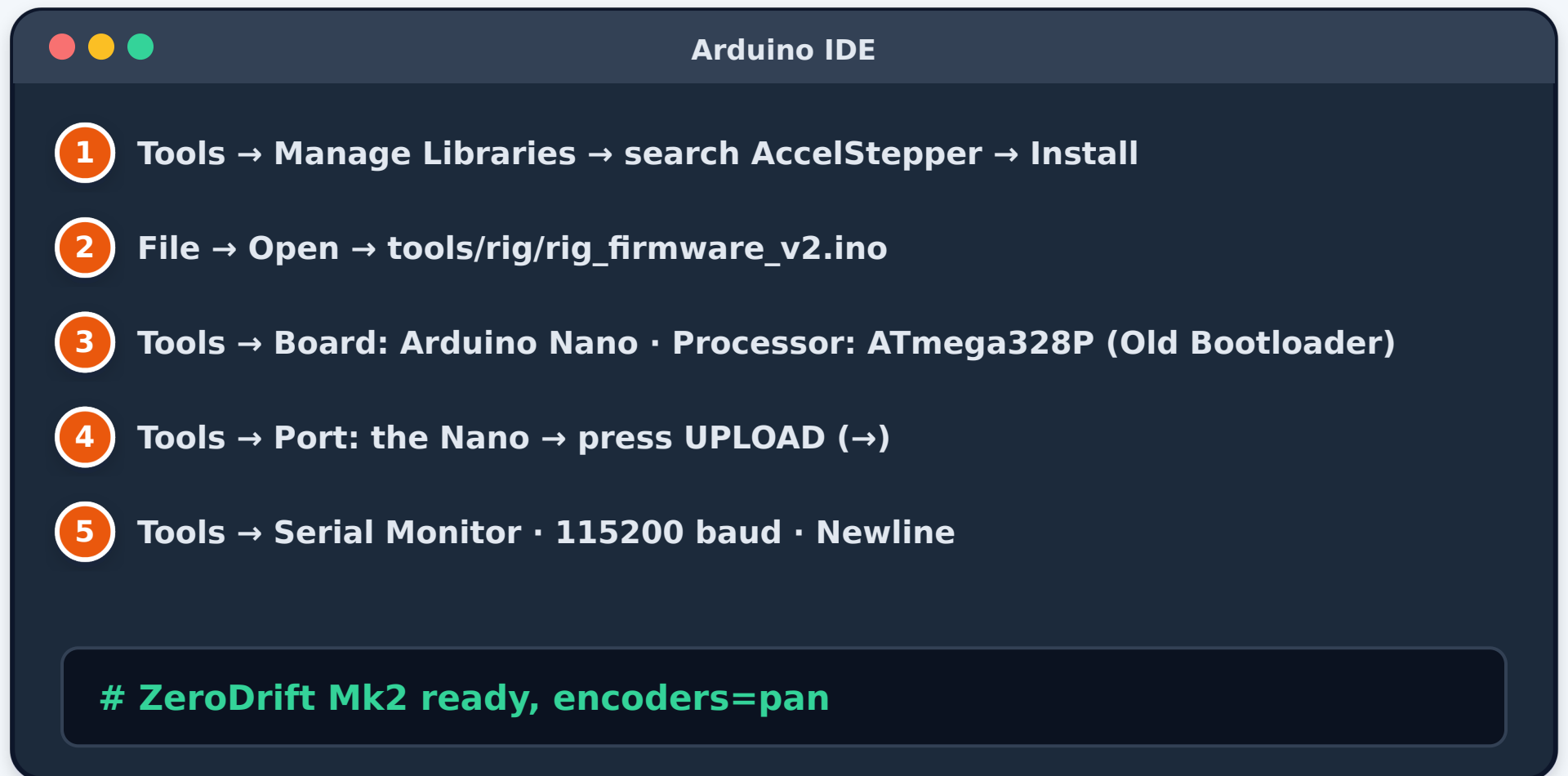
MAKE THEM LOOK THE SAME



- 1 Battery + → switch → **100 Ω** → LED long leg. LED short leg → battery –.
- 2 Same kind of box and the other half of the ping-pong ball, so it looks exactly like the beacon.
- 3 It stays ON and **never blinks**. The rig must ignore it and keep watching the beacon.

✓ Switch on: a steady red light. No light? Flip the LED round.

Put the code in the Arduino



TYPE THESE TO TEST (press Enter after each)

P267 T0

bottom motor turns 30°

P0 T0

back to the start

T89

top motor nods 10°

L1 · L0

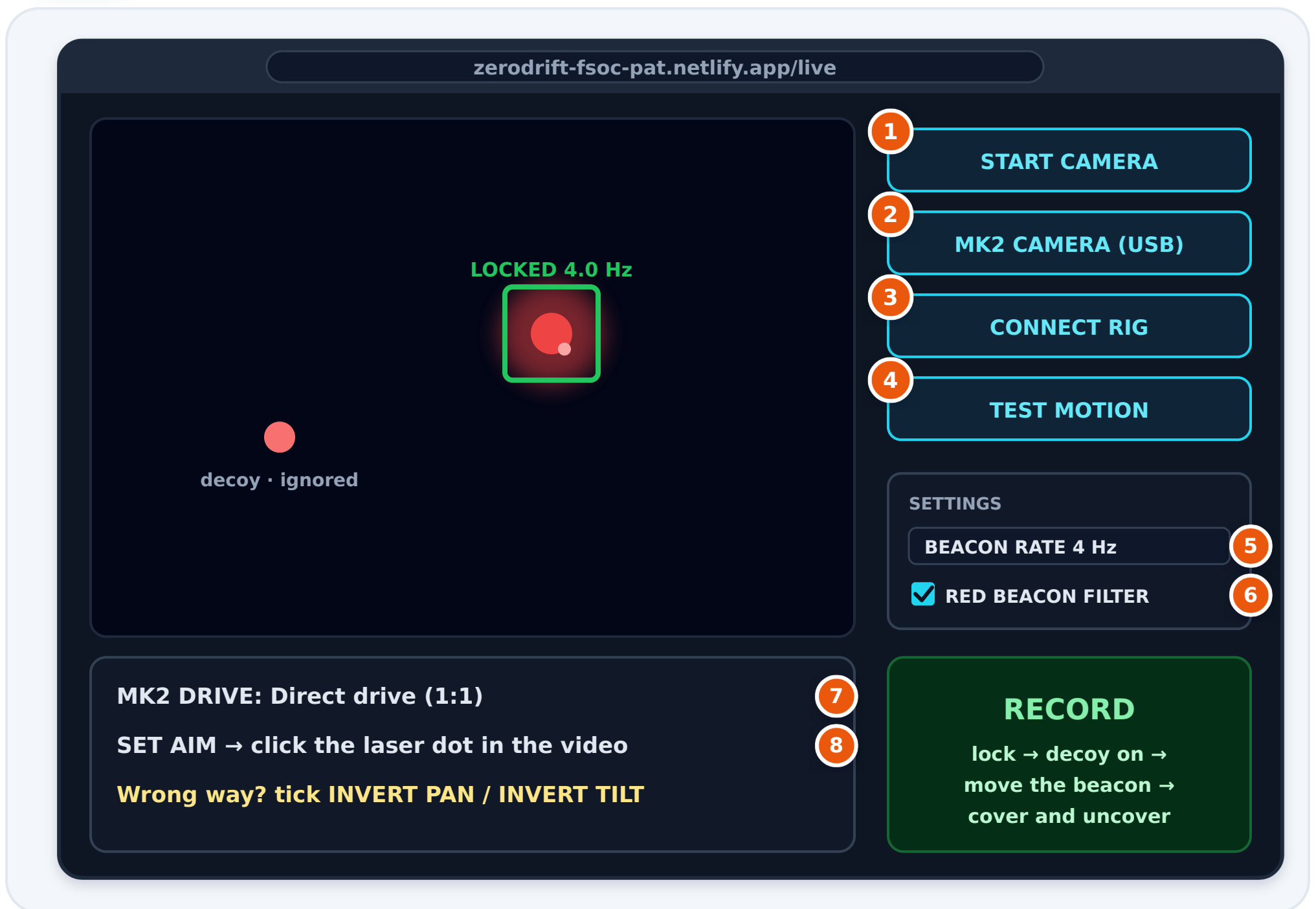
laser on · off

- 1 Plug the rig Arduino into the laptop. Do steps 1–5 on the screen above.
- 2 If upload fails, choose the other Processor option and try again.
- 3 Type the test lines. Then **close the Serial Monitor** — the website can't connect while it is open.



You see "ZeroDrift Mk2 ready" and the motors move when you type P267 T0.

Go live on the website



- 1 Chrome → the **/live** page → **START CAMERA** → **MK2 CAMERA (USB)**. No webcam yet? The laptop camera works for testing.
- 2 **CONNECT RIG** → pick the Arduino → **TEST MOTION**. Wrong way? Tick INVERT.
- 3 Beacon ON. Red beacon in a bright room: tick **RED BEACON FILTER**. Then SET AIM on the laser dot.



It locks in about a second and turns to face the beacon. Now record the demo!